BANDA SRI HARI

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GitHub | LinkedIn | Portfolio

EDUCATION

Sir M. Visvesvaraya Institute of Technology, Bengaluru — **Expected 2027**
B.E. in Artificial Intelligence & Machine Learning — **CGPA: 8.72/10**

EXPERIENCE

Chiacon Consulting Private Limited — **Mar 2026 – Jun 2026**
AI Engineer Intern — AI Voice Interview & HR Orchestration Platform — Remote, India

- Engineered a real-time **Python/FastAPI WebSocket** voice system integrating Recall.ai and Azure Speech with continuous audio streaming and thread-safe speech-SDK callbacks.
- Architected a **9-node LangGraph** workflow with typed state, deterministic routing, project deduplication, guardrails, and bounded memory for reliable multi-turn control.
- Built concurrent **Technical, Behavioral, and Integrity LLM evaluators** using `asyncio.gather()`, Pydantic schemas, and OpenAI Function Calling for structured assessment outputs.
- Integrated **Azure OpenAI, Microsoft Graph, Microsoft Entra ID, Recall.ai, and Azure Communication Services** through REST APIs using OAuth 2.0/HMAC, including a raw REST workaround for an SDK limitation.
- Developed a headless **7-persona regression harness** with LLM simulation, mocked audio dependencies, and 5 assertions; debugged state-machine, SDK, acoustic-feedback, and file-write failures.

SKILLS

Languages: Python, SQL, JavaScript
AI / GenAI: LLMs, Generative AI, RAG, Embeddings, Vector Search, AI Agents, AI Orchestration, LangChain, LangGraph, Structured Outputs, Function Calling, Prompt Engineering, Context Engineering, Guardrails, MCP
Azure & Cloud: Azure AI Foundry, Azure OpenAI, Azure Speech, Microsoft Graph, Microsoft Entra ID, Azure Communication Services, AWS Lambda, API Gateway, S3, Amazon Bedrock, Bedrock Knowledge Bases, Docker
Backend & APIs: FastAPI, REST APIs, WebSockets, Webhooks, OAuth 2.0, Pydantic, Async Programming, Concurrency
Databases & Retrieval: PostgreSQL, pgvector, FAISS, SQLite, DynamoDB, Hybrid Retrieval, Tree-sitter/AST
Automation & Reliability: GitHub Actions, Pytest, Regression Testing, Mocking, LLM Evaluation, Evaluation Harnesses, Pipeline Tracing, Debugging

PROJECTS

GitNova – Repository-Aware Code Intelligence Platform — Live | GitHub

- Built and deployed a full-stack **repository-aware code intelligence platform** with Python/FastAPI, PostgreSQL/pgvector, React, and GitHub REST ingestion to help developers investigate actionable issues in unfamiliar codebases.
- Engineered repository-grounded retrieval over **41,000+ AST-indexed code chunks** using code embeddings, pgvector similarity search, PostgreSQL full-text search, Tree-sitter/AST parsing, and Reciprocal Rank Fusion.
- Designed a dual-phase **Gemini/Groq reasoning pipeline** with structured Pydantic outputs and a fail-closed `GroundingVerif` validating files, symbols, test commands, and repository evidence before publication.
- Built **automated tests and rolling RAG evaluation** against merged PR diffs with stage-level tracing for latency/model tracking; resolved pgBouncer, PostgREST RPC, and cloud build-memory failures.

Stack: Python, FastAPI, PostgreSQL, pgvector, React, Tree-sitter, Gemini, Groq, GitHub Actions, Docker

Orbit – Serverless AI Learning Platform — GitHub

- Architected and deployed a serverless **Python/FastAPI application on AWS Lambda/Mangum**, exposing 8 API Gateway endpoints for code evaluation, viva, session, curriculum, and mentor workflows.
- Engineered **DynamoDB assessment state** with composite keys, atomic updates, and state-machine transitions for attempts, viva locking, and learning progression without relying on client-side state.
- Built managed **RAG with S3 and Bedrock Knowledge Bases**, ingesting timestamped YouTube lecture transcripts for semantic retrieval and grounded mentor responses.
- Integrated **Amazon Bedrock and Guardrails** for code evaluation, viva generation/grading, and mentor interactions; resolved API Gateway timeouts, Guardrail false positives, and non-JSON failures with defensive fallbacks.

Stack: Python, FastAPI, AWS Lambda, Mangum, API Gateway, DynamoDB, S3, Amazon Bedrock

Unshell – AML & KYB Forensic Intelligence Graph — *Hackfest 2026 – 2nd Runner-Up* | GitHub

- Built an **AML/KYB forensic intelligence system** using Python/FastAPI, Companies House REST data, PDF RAG, deterministic NetworkX analysis, and OFAC SDN screening to trace hidden ownership.
- Engineered a deterministic **NetworkX risk engine** using Johnson's cycle detection, degree/path-depth checks, and 11 rule-based typologies for circular ownership, nominee pass-throughs, and layered structures.
- Designed a fail-closed **grounding pipeline** with PyMuPDF, `all-MiniLM-L6-v2`, FAISS, Gemini structured outputs, and RapidFuzz to validate entities and evidence before graph analysis.
- Built resilient multi-source ingestion with recursive Level-2 Companies House traversal, **FastMCP data-broker integration**, rate-limit handling, and indexed SQLite/`aiosqlite` OFAC SDN screening.

Stack: Python, FastAPI, LangGraph, PyMuPDF, FAISS, Sentence-Transformers, Gemini, RapidFuzz, NetworkX, SQLite, FastMCP

ACHIEVEMENTS & CERTIFICATIONS

- **Hackfest 2026:** 2nd Runner-Up – Top 50 / 750+ teams
- **Amazon ML Challenge 2025:** Top 15% among 10,000+ teams
- **NPTEL:** Mathematical Foundations for Machine Learning – Elite Silver (IIT Guwahati)
- **Oracle Certified:** Generative AI Professional, Data Science Professional, Database@AWS Foundations Associate